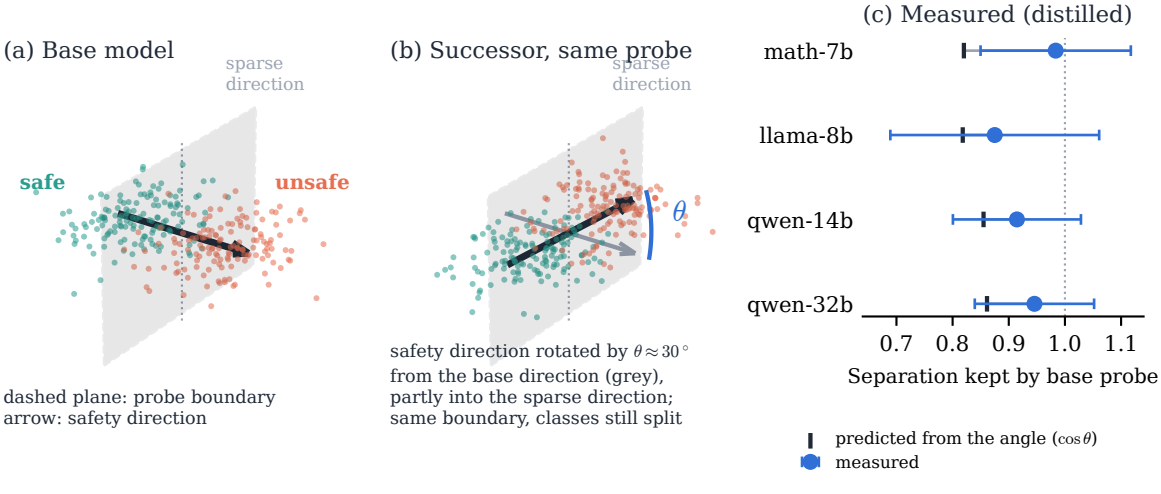
(a) Base model
sparse direction
safe
unsafe
dashed plane: probe boundary
arrow: safety direction
(b) Successor, same probe
sparse direction
θ
safety direction rotated by θ ≈ 30°
from the base direction (grey),
partly into the sparse direction;
same boundary, classes still split
(c) Measured (distilled)
math-7b
llama-8b
qwen-14b
qwen-32b
0.7
0.8
0.9
1.0
1.1
Separation kept by base probe
predicted from the angle (cos θ)
measured